Contribution Journal Derrick K. Gibbs-McGlaston

My main contribution to the Module 05 Data Preparation lab was completing and reviewing the data preprocessing steps in the notebook.

I worked through the required tasks and checked the results after each step. This included handling the missing values in the Age column, converting the categorical Sex and Embarked information using one-hot encoding, and standardizing the Age and Fare features.

My focus was making sure each preparation step worked correctly before moving forward. I reviewed how the dataset changed after each operation instead of only checking whether the code ran without an error.

This contribution helped make sure the data preparation portion of the notebook was completed correctly and that the resulting data was in a better form for the next stages of the machine learning workflow.

Marlon Roberts- my contribution to this lab was creating the GitHub for my team to access, I reviewed the data and completed the word needed for the module. I copy and paste all work into one file and submit it, I make sure the work is finalized. I edit the document if needed, and submit.

Lab 05 – Data Preparation

Individual Contribution Journal
Student: Victor Djuma
Course: ITAI 1371
Assignment: Module 05 – Data Preparation
For Lab 05, my contribution focused on completing the Data Preparation notebook in Google Colab and making sure each required task was completed correctly. I began by running the setup section, which loaded the Titanic dataset and displayed the missing values. This helped me identify that the dataset contained missing values, especially in the Age column, which was one of the main problems addressed in the lab. The notebook explained that raw data often requires preprocessing before it can be used by a machine learning model. Module 05 Lab - Data Preparation
For the first task, I worked on **handling missing values** in the Age column. I calculated the median age and used it to replace the missing Age values. After completing the imputation, I verified the result by checking the number of missing values remaining in the Age column, which became zero. This task helped me understand how imputation can

preserve useful data instead of simply deleting incomplete rows. Module 05 Lab - Data Preparation
For the second task, I completed the **One-Hot Encoding** section. I used pd.get_dummies() to convert the categorical Sex and Embarked columns into numerical columns. I also used drop_first=True as directed in the notebook. After running the code, I checked the new DataFrame and confirmed that the original categorical columns had been replaced with encoded columns such as Sex_male, Embarked_Q, and Embarked_S. Module 05 Lab - Data Preparation
For the third task, I completed the **feature scaling** section. I used StandardScaler from scikit-learn to scale the Age and Fare columns. After running the code, I reviewed the transformed values and confirmed that Age and Fare were standardized. This helped me understand why numerical features on different scales may need to be transformed before being used by some machine learning algorithms. Module 05 Lab - Data Preparation
I also completed the three **Knowledge Check** questions at the end of the notebook. I explained why the median is useful for imputation, how One-Hot Encoding converts categorical information into numerical form, and why Decision Tree models generally do not require feature scaling. Module 05 Lab - Data Preparation
Finally, I reviewed the notebook to make sure the required code cells executed successfully, saved the completed Jupyter notebook, and created a PDF version of the lab. My contribution to Lab 05 included completing the required coding tasks, checking the outputs, answering the Knowledge Check questions, and preparing my completed work for submission.

Lab 5-Data Preparation
ITAI 1371
Team: ThetaBetas
Contribution Journal
Brandy Gonzales
I completed the Lab 5 Data Preparation notebook on the Titanic dataset. I filled the 177 missing
Age values with the median and checked that none were left. I used one-hot encoding to turn
Sex and Embarked into number columns, and I scaled Age and Fare with StandardScaler. I also
answered all three Knowledge Check questions in my own words: median vs. mean, what one-hot encoding does and why, and why a decision tree doesn't need scaling. Finally, I wrote
my reflection on what I learned, saved as a PDF, and sent my work to our team leader Marlon Roberts, who then submits on our behalf.

Lab 05 Contribution Journal: Data Preparation
Pablo Lara, ITAI 1371

For this lab, each team member completed the full notebook on their own in Google Colab,
and then we combined our work and journals into one submission. My contributions were:

Completed the full notebook on my own. I worked through all three coding tasks in Colab:

- Task 1, Imputation: I calculated the median of the Age column (28.0) and used it to fill in the 177 missing ages. I then checked that no missing values remained.
- Task 2, One-Hot Encoding: I encoded the Sex and Embarked columns using pd.get_dummies() with drop_first=True. Sex became a single Sex_male column, and Embarked became Embarked_Q and Embarked_S.
- Task 3, Feature Scaling: I used StandardScaler to standardize the Age and Fare columns so they have a mean of 0 and a standard deviation of 1.
- Knowledge Check: I answered the three questions on median vs. mean, one-hot encoding, and whether Decision Trees need scaling.

Found and fixed a problem with the notebook file. The notebook had lost its line breaks, so every code cell was squashed onto one line and most cells gave a syntax error. I fixed each cell by putting every statement on its own line.

Caught a silent error and warned the team. The final scaling cell ran without any error but did not actually scale anything, because everything after the first # was treated as a comment. I noticed something was wrong, corrected the cell, and let my team know to watch for this issue in the last code cell, including how to check their output to confirm the
scaling worked.

Verified my results. I re-ran the cells to make sure my outputs were correct and that the notebook worked from start to finish.